

**NIRMA UNIVERSITY**  
**Institute of Technology**  
**B.Tech. Computer Science and Engineering/ BS/ Int BTech MBA**  
**Practical List**  
**Semester-V**

|                   |                               |                  |
|-------------------|-------------------------------|------------------|
| <b>3CS524CC26</b> | <b>Full Stack Development</b> | <b>[3-0-2-4]</b> |
|-------------------|-------------------------------|------------------|

| <b>Sr. No.</b> | <b>Practical Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Hours</b> | <b>CLO</b> |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| 1              | I. Demonstrate JavaScript code embedded within the HTML file<br>II. Install Node.js: Download and install Node.js from nodejs.org.<br>III. create a “script.js” file to print “Hello world” on the server side<br>IV. create an “app.js” file to create and store JSON Object in the File<br>V. Identify the application title that will be used to perform Practicals 2 to 10                                                                                                                                                                                                                                  | 02           | 1          |
| 2              | I. Create a JSON object and display the same using node.js in the terminal.<br>II. Create a Node.js script to read and display the content of the external JSON file.<br>III. Create and display multi-dimensional JSON Arrays, also demonstrate Accessing individual elements of the multi-dimensional JSON Arrays.<br>IV. Create a web-based application example using <b>JavaScript</b> and <b>JSON</b> to manipulate JSON data. In this example, create a simple to-do list application where tasks are stored in a JSON array. Users can add tasks, mark them as completed, and remove them from the list. | 02           | 1          |
| 3              | I. Create a responsive web page involves using the viewport <u>meta</u> tag in the head of your HTML document.<br>II. Create a responsive web page by using the width and max-width Properties and Changing <u>images</u> according to the browser width.<br>III. Create a responsive web page that demonstrates responsive text size using the "vw" (viewport width) unit in CSS<br>IV. Create a responsive web page that demonstrates <u>media queries</u> for creating breakpoints in the web page.                                                                                                          | 02           | 3          |
| 4              | I. Implement <b>Geolocation API</b> to retrieve the user's current location.<br>II. Implement JavaScript to interact with the <b>Local Storage</b><br>III. Demonstrating the <b>Drag and Drop API</b>                                                                                                                                                                                                                                                                                                                                                                                                           | 02           | 2          |
| 5              | I. Demonstrate the implementation of Bootstrap online (using the <b>CDN link</b> ).<br>II. Demonstrate the implementation of Bootstrap offline.<br>III. Create a simple web page using Bootstrap 5 with HTML for student registration The web page must include the title of the page, roll number, first name, mobile number, email ID, and address.                                                                                                                                                                                                                                                           | 02           | 3          |
| 6              | Demonstrate Tailwind CSS through HTML web pages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 04           | 3          |
| 7              | Demonstrate how to create a custom directive and how to use it through HTML web pages (Vue JS).<br>I. Create a custom directive to transform text to uppercase when clicked.<br>II. Creating a Dynamic List with Vue.js Custom Directive<br>III. Create a directive that formats and displays dates in a human-readable format                                                                                                                                                                                                                                                                                  | 04           | 4          |
| 8              | I. Demonstrate Node.js using MySQL to perform the below operations: Create a database, create a table, insert a record, and update the record.<br>II. Demonstrate Node.js using MySQL to perform the below operations using the user-defined function: delete record, select record, select unique, and drop table.                                                                                                                                                                                                                                                                                             | 04           | 1          |
| 9              | Demonstrate a Role-Based Authentication and Access Control System using JavaScript.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 04           | 4          |
| 10             | Demonstrate Node.js using MongoDB to perform Create, Read, Update, and Delete operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 04           | 4          |
| 11*            | Deploy the application and manage version control using Git and GitHub.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -            | 4          |
| <b>Total</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30           |            |